Question 31

Correct

Mark 10.00 out of 10.00

Remove flag

WAP to check if a number is an Abundant number , A number n is said to be Abundant Number to follow these condition

- The sum of its proper divisors is greater than the number itself
- The difference between these two values is called the abundance.

For example:

Test	Input	Result
case 1	18	18 is an Abundant Number Num: 18 Sum: 21 Abundance: 3
case 2	20	20 is an Abundant Number Num: 20 Sum: 22 Abundance: 2
case 3	10	10 is not a Abundant Number

Answer: (penalty regime: 0 %)

Reset answer

```c
#include <stdio.h>

int main ()
{
    //logic
    double num;
    double sum = 0;
    scanf("%lg",&num);
    for(int i=1;i<num;i++){
        if( (int)num % i == 0)
            sum = sum + i;
    }
    if(sum>num){
        printf("%lg is an Abundant Number\n",num);
        printf("Num: %lg\n",num);
        printf("Sum: %lg\n",sum);
        printf("Abundance: %g",sum-num);
    }else{
        printf("%lg is not a Abundant Number",num);
    }
    return 0;
}
```

Question 32

Correct

Mark 10.00 out of 10.00

Flag question

Write a program in c to count a total number of duplicate elements in an array

For example:

Test	Input	Result
case 1	3 5 1 1	Total number of duplicate elements found in the array : 1
case 2	5 7 7 5 5 15	Total number of duplicate elements found in the array : 2

Answer: (penalty regime: 0 %)

Reset answer

```c
#include <stdio_ext.h>


// Main function
int main() {
    //Read the size from user
    int size;
    scanf("%d",&size);

    //Read elements from user
    int arr[size];
    for(int i=0;i<size;i++){
        scanf("%d",&arr[i]);
    }
    //Define your logic here
    int count = 0;
    for(int i=0;i<size;i++){
        for(int j=i+1;j<size;j++){
            if(arr[i]==arr[j]){
                count++;
                break;
            }
        }
    }
    printf("Total number of duplicate elements found in the array : %d",count);

    return 0;
}

```

	Test	Input	Expected	Got	
✓	case 1	3 5 1 1	Total number of duplicate elements found in the array : 1	Total number of duplicate elements found in the array : 1	✓
✓	case 2	5 7 7 5 5 15	Total number of duplicate elements found in the array : 2	Total number of duplicate elements found in the array : 2	✓

Question 33

Partially correct

Mark 4.00 out of 10.00

Flag question

Write a function to count the digits of a given number using recursion.

For example:

Test	Result
`printf("Number of digits: %d", countDigits(50));`	`Number of digits: 2`
`printf("Number of digits: %d", countDigits(100));`	`Number of digits: 3`

Answer: (penalty regime: 0 %)

Reset answer

```c
#include <stdio.h>


int countDigits(int num)
{
    // Recursive function to count the digits of a number
    static int count=0;
    if(num==0)
        return 1;
    if(num==1)
        return 1;

    countDigits(num/10);
    count++;
    return count;
}

```

	Test	Expected	Got	
✓	`printf("Number of digits: %d", countDigits(50));`	`Number of digits: 2`	`Number of digits: 2`	✓
✗	`printf("Number of digits: %d", countDigits(100));`	`Number of digits: 3`	`Number of digits: 4`	✗
✓	`printf("Number of digits: %d", countDigits(0));`	`Number of digits: 1`	`Number of digits: 1`	✓

Some hidden test cases failed, too.

Show differences

▸ **Show/hide question author's solution (C)**

Partially correct

Question 34

Correct

Mark 10.00 out of 10.00

Flag question

Write a function to sort any given string in alphabetical order.

For example:

Test	Result

`char str[100] = "boat";`
`bubbleSort(str);`
`printf("Sorted string: %s\n", str);`
	`Sorted string: abot`

`char str[100] = "bangalore";`
`bubbleSort(str);`
`printf("Sorted string: %s\n", str);`
	`Sorted string: aabeglnor`

Answer: (penalty regime: 0 %)

Reset answer

```c
#include <stdio.h>
#include <string.h>
#include <stdlib.h>

void bubbleSort(char arr[])
{
    //Define your logic here
    int i=0;
    char temp;
    while(arr[i] != '\0'){
        int j = i+1;
        while(arr[j] !='\0'){
            if(arr[i] >= arr[j]){
                temp = arr[i];
                arr[i] = arr[j];
                arr[j] = temp;
            }
            j++;
        }
        i++;
    }
}


```

	Test	Expected	Got	

✓	`char str[100] = "boat";`
	`bubbleSort(str);`
	`printf("Sorted string: %s\n", str);`
		`Sorted string: abot`	`Sorted string: abot`	✓

Question 35

Partially correct

Mark 2.00 out of 10.00

Flag question

write a program to print a decimal number in binary equivalent and reverse of a decimal number using recursion.

For example:

Test	Input	Result
case 1	10	Binary equivalent: 1010 Reversed number: 1

Question 36

Incorrect

Mark 0.00 out of 10.00

Flag question

Write a program to reverse the elements in each 1D array.

For example:

Test	Input	Result
case 1	3 3 4 5 6 7 8 9 2 3 4	Before reverse: 4 5 6 7 8 9 2 3 4 Reversed matrix: 6 5 4 9 8 7 4 3 2
case 2	2 2 4 3 6 5	Before reverse: 4 3 6 5 Reversed matrix: 3 4 5 6

Answer: (penalty regime: 0 %)

Reset answer

Question 37

Not answered

Mark 0.00 out of 10.00

Flag question

Write a program to read and print an employee's detail using structure

For example:

Test	Input	Result
case 1	100 Ramesh Bangalore 10000 Married	Employee details are : Id - 100 Name - Ramesh Address - Bangalore Salary - 10000 Marital Status - Married
case 2	200 Dinesh Chennai 25000 Married	Employee details are : Id - 200 Name - Dinesh Address - Chennai Salary - 25000 Marital Status - Married